2025 WSMO Team Round Solutions

SMO Team

TR 1: Define $f(n)$ as the remainder when n is divided by 5. Compute

$$f(1) + f(2) + f(3) + \cdots + f(2025).$$

Answer: $\boxed{4050}$

Solution: Note that $f(n)$ cycles through $1, 2, 3, 4, 0$, which sum to 10. Since $2025 = 5 \cdot 405$, the sum consists of 405 complete cycles, giving an answer of $405 \cdot 10 = \boxed{4050}$.

TR 2: Let a and b be positive reals such that $\frac{1}{a} + \frac{1}{b} = \frac{1}{6}$. Find the minimum possible value of $a + b$.

Answer: $\boxed{24}$

Solution: From the Cauchy-Schwarz Inequality, we have

$$(a+b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq (1+1)^2 = 4 \implies \frac{a+b}{6} \geq 4 \implies a + b \geq 24,$$

with equality when $a = b = 12$. So, the answer is $\boxed{24}$.

TR 3: How many distinct ways are there to arrange the letters in "SOLSTICE" such that no two vowels are adjacent and the resulting word does not begin or end with a vowel?

Answer: $\boxed{1440}$

Solution: The letters in "SOLSTICE" consist of the vowels O, I, and E and the consonants S, L, S, T, and C. There are $\frac{5!}{2!} = 60$ ways to arrange the consonants. Since no two vowels are adjacent and no vowel is at either end, the vowels must be placed in 3 of the 4 gaps between consecutive consonants, giving $4 \cdot 3 \cdot 2 = 24$ ways. So, the answer is $60 \cdot 24 = \boxed{1440}$.

TR 4: For how many positive integers $n \leq 2025$ does n^6 divide evenly into 6^n?

Answer: $\boxed{41}$

Solution: Since $n^6 \mid 6^n$, the only prime factors of n are 2 and 3. Let $n = 2^a 3^b$. Since $v_2(6^n) = v_3(6^n) = n$, we need

$$6a \leq n \quad \text{and} \quad 6b \leq n.$$

First, we count the integers of the form $2^a 3^b$ that are at most 2025. For $b = 0, 1, 2, 3, 4, 5, 6$, there are $11, 10, 8, 7, 5, 4, 2$ possible values of a, respectively, for a total of 47. Now, if $a \geq 5$,

then $n \geq 2^a \geq 6a$ and $n \geq 32 \geq 6b$ for $b \leq 5$, while if $b \geq 3$, then $n \geq 3^b \geq 6b$ and $n \geq 27 \geq 6a$ for $a \leq 4$. So, we only need to check $a \leq 4$ and $b \leq 2$, where the integers that fail are $2, 3, 4, 8, 9$, and 16. Thus, the answer is $47 - 6 = \boxed{41}$.

TR 5: Anna picks two distinct prime numbers $p > q$ such that $n = pq(p - q) < 1000$. She tells Bob the value of n, but Bob is not able to uniquely determine the values of p and q. Find the largest possible value of n.

Answer: $\boxed{646}$

Solution: Note that if $p - q$ is also prime, then the pairs (p, q) and $(p, p - q)$ give the same value of n. Since q and $p - q$ sum to the odd prime p, one of them must be 2, so this happens exactly for the pairs $(p, 2)$ and $(p, p - 2)$ when p and $p - 2$ are both prime. Now, we list all values of $n \geq 646$. For each prime q, we need $pq(p - q) < 1000$, giving

$$\begin{aligned} q = 2: &\quad 2 \cdot 19 \cdot 17 = 646,\ 2 \cdot 23 \cdot 21 = 966, \\ q = 3: &\quad 3 \cdot 17 \cdot 14 = 714,\ 3 \cdot 19 \cdot 16 = 912, \\ q = 13: &\quad 13 \cdot 17 \cdot 4 = 884, \\ q = 17: &\quad 17 \cdot 19 \cdot 2 = 646, \end{aligned}$$

and all other pairs give $n < 646$. Checking the prime factorizations of 714, 884, 912, and 966, each of these comes from exactly one pair (p, q). However, 646 comes from both $(19, 2)$ and $(19, 17)$, so the answer is $\boxed{646}$.

TR 6: Let $d(n)$ be a function from the positive integers to the integers defined by

$$d(n) = \sum_{m>0 \text{ divides } n} (-1)^{d(m)}.$$

Find the value of $d(30^9)$.

Answer: $\boxed{750}$

Solution: Let $\tau(n)$ denote the number of divisors of n and $s(n)$ denote the number of perfect square divisors of n. We claim that $d(n) = \tau(n) - 2s(n)$, and in particular, $d(n)$ is odd if and only if n is a perfect square. We proceed by strong induction on n. Suppose the claim holds for all proper divisors of n. Then,

$$d(n) - (-1)^{d(n)} = \sum_{m|n,\ m<n} (-1)^{d(m)}.$$

If n is a perfect square, then $\tau(n)$ is odd, and the right hand side is $(\tau(n) - 1) - 2(s(n) - 1)$, which is even. This forces $d(n)$ to be odd, so $d(n) = \tau(n) - 2s(n)$. Otherwise, $\tau(n)$ is even, and the right hand side is $(\tau(n) - 1) - 2s(n)$, which is odd. This forces $d(n)$ to be even, so $d(n) = \tau(n) - 2s(n)$ again. Now, $30^9 = 2^9 \cdot 3^9 \cdot 5^9$, so $\tau(30^9) = 10^3 = 1000$ and $s(30^9) = 5^3 = 125$. So, the answer is $1000 - 2 \cdot 125 = \boxed{750}$.

TR 7: Let $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ be sequences of positive real numbers with the property that for every positive integer n,

$$a_{n+1}a_n = b_{n+1}b_n + b_{n+1}a_n + a_{n+1}b_n.$$

Given that $a_1 = 25$, $b_1 = 20$, and $a_{100} = 2025$, find the value of b_{100}.

Answer: $\boxed{225}$

Solution: Adding $a_{n+1}a_n$ to both sides, we have

$$2a_{n+1}a_n = a_{n+1}a_n + a_{n+1}b_n + b_{n+1}a_n + b_{n+1}b_n = (a_{n+1} + b_{n+1})(a_n + b_n).$$

Let $c_n = \frac{a_n + b_n}{a_n}$, so $c_{n+1}c_n = 2$ for all n. This means $c_{n+2} = c_n$, so

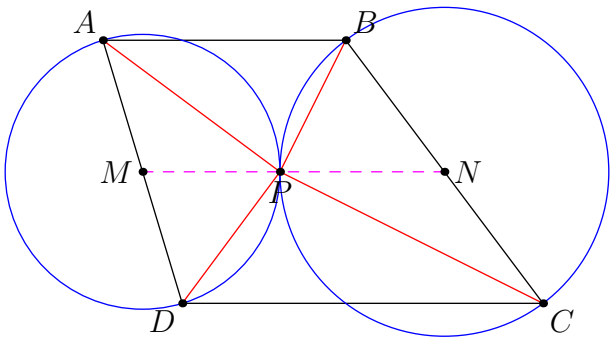
$$c_{100} = c_2 = \frac{2}{c_1} = \frac{2}{\frac{45}{25}} = \frac{10}{9}.$$

Thus, $a_{100} + b_{100} = \frac{10}{9} \cdot 2025 = 2250$, meaning $b_{100} = 2250 - 2025 = \boxed{225}$.

TR 8: Let $ABCD$ be a trapezoid with $AB \parallel CD$. Circles ω_1 and ω_2 have diameters AD and BC, respectively, and are tangent at point P. Given that $PA = 3$, $PB = 2$, and $PC = 4$, the value of PD^2 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{173}$

Solution:



We will use analytic geometry. Let M and N be the midpoints of AD and BC, which are the centers of ω_1 and ω_2. Since ω_1 and ω_2 are tangent at P, P lies on line MN, which is parallel to AB and CD. Let $P = (0, 0)$ and let MN be the x-axis, so $A = (a, t)$, $B = (b, t)$, $C = (c, -t)$, and $D = (d, -t)$ for some $t > 0$. Since AD and BC are diameters, $\angle APD = \angle BPC = 90°$, meaning

$$ad - t^2 = 0 \quad \text{and} \quad bc - t^2 = 0.$$

From $PB = 2$ and $PC = 4$, we have $b^2 = 4 - t^2$ and $c^2 = 16 - t^2$, so

$$(4 - t^2)(16 - t^2) = b^2c^2 = t^4 \implies 64 - 20t^2 = 0 \implies t^2 = \frac{16}{5}.$$

From $PA = 3$, we have $a^2 = 9 - t^2 = \frac{29}{5}$, so $d^2 = \frac{t^4}{a^2} = \frac{256}{145}$. Thus,

$$PD^2 = d^2 + t^2 = \frac{256}{145} + \frac{16}{5} = \frac{144}{29} \implies 144 + 29 = \boxed{173}.$$

TR 9: Let k be a positive integer such that the equation

$$\left\lfloor \frac{k}{\left\lfloor \frac{k}{n} \right\rfloor} \right\rfloor = n$$

has exactly 25 positive integer solutions n. Find the largest possible value of k.

Answer: $\boxed{181}$

Solution: Note that $n \leq k$, since otherwise $\left\lfloor \frac{k}{n} \right\rfloor = 0$. Let $q = \left\lfloor \frac{k}{n} \right\rfloor$. The integers m with $\left\lfloor \frac{k}{m} \right\rfloor = q$ are exactly those with $\frac{k}{q+1} < m \leq \frac{k}{q}$, and the largest of them is $\left\lfloor \frac{k}{q} \right\rfloor$. So, n is a solution if and only if it is the largest m with $\left\lfloor \frac{k}{m} \right\rfloor = q$, meaning the number of solutions is the number of distinct values of $\left\lfloor \frac{k}{n} \right\rfloor$ for $1 \leq n \leq k$. Let $s = \left\lfloor \sqrt{k} \right\rfloor$. For $1 \leq n \leq s$, the values $\left\lfloor \frac{k}{n} \right\rfloor$ are at least s and are distinct, since

$$\frac{k}{n} - \frac{k}{n+1} = \frac{k}{n(n+1)} \geq 1$$

for $n \leq s-1$. For $n \geq s+1$, the values $\left\lfloor \frac{k}{n} \right\rfloor$ are exactly $1, 2, \ldots, \left\lfloor \frac{k}{s+1} \right\rfloor$, which is at most s. If $k \geq s(s+1)$, then $\left\lfloor \frac{k}{s} \right\rfloor \geq s+1$ and $\left\lfloor \frac{k}{s+1} \right\rfloor = s$, giving $2s$ distinct values. Otherwise, $\left\lfloor \frac{k}{s+1} \right\rfloor = s-1$, giving $2s-1$ distinct values. Since 25 is odd, we need $2s-1 = 25$, so $s = 13$ and $k < 13 \cdot 14 = 182$. So, the largest possible value of k is $\boxed{181}$.

TR 10: Let a, b, and c be complex numbers such that $|a| = |b| = |c| = 1$. Suppose that the three summations

$$\sum_{i=0}^{5} a^i b^{5-i}, \ \sum_{i=0}^{7} b^i c^{7-i}, \text{ and } \sum_{i=0}^{11} c^i a^{11-i}$$

are all nonzero real numbers. Find the number of possible ordered triples (a, b, c).

Answer: $\boxed{770}$

Solution: Let $S = \sum_{i=0}^{5} a^i b^{5-i}$. Since $|a| = |b| = 1$, we have $\overline{a} = \frac{1}{a}$ and $\overline{b} = \frac{1}{b}$, so

$$\overline{S} = \sum_{i=0}^{5} a^{-i} b^{i-5} = \frac{1}{(ab)^5} \sum_{i=0}^{5} a^{5-i} b^i = \frac{S}{(ab)^5}.$$

So, if $S \neq 0$, then S is real if and only if $(ab)^5 = 1$. In the same manner, we need $(bc)^7 = 1$ and $(ca)^{11} = 1$. There are $5 \cdot 7 \cdot 11 = 385$ ways to choose the values of ab, bc, and ca. Given these values, $(abc)^2 = (ab)(bc)(ca)$ gives 2 possible values of abc, which determines $a = \frac{abc}{bc}$, $b = \frac{abc}{ca}$, and $c = \frac{abc}{ab}$. This gives 770 ordered triples. Now, we check that the sums are nonzero. Note that $a^2 = \frac{(ab)(ca)}{bc}$, so $\left(a^2\right)^{385} = 1$. Since $(ab)^{385} = 1$ as well, we have

$$\left(\frac{a}{b}\right)^{385} = \frac{\left(a^2\right)^{385}}{(ab)^{385}} = 1.$$

If $S = 0$, then $a \neq b$ and $a^6 = b^6$, so $\left(\frac{a}{b}\right)^6 = 1$. Since $\gcd(6, 385) = 1$, this forces $a = b$, a contradiction. In the same manner, $\left(\frac{b}{c}\right)^{385} = \left(\frac{c}{a}\right)^{385} = 1$, and since $\gcd(8, 385) = \gcd(12, 385) = 1$, the other two sums are also nonzero. So, the answer is $\boxed{770}$.

TR 11: Let A, B, and C be collinear in that order such that $AB = 7$ and $AC = 17$. Additionally, let B, M, and N be collinear in that order such that $BN = 10$ and $MN = 3$. Let P be a point such that $NP \perp AN$ and $NP = MC$. As M and N vary, the points P trace out a closed curve that encloses a region with area $k\pi$. Find k.

Answer: $\boxed{298}$

Solution: We will use analytic geometry. Let $B = (0, 0)$, $A = (-7, 0)$, and $C = (10, 0)$. Since $BN = 10$ and $BM = 7$, we have $N = (10\cos\theta, 10\sin\theta)$ and $M = (7\cos\theta, 7\sin\theta)$ for some angle θ. So,

$$AN^2 = (10\cos\theta + 7)^2 + (10\sin\theta)^2 = 149 + 140\cos\theta, \quad \text{and}$$
$$MC^2 = (7\cos\theta - 10)^2 + (7\sin\theta)^2 = 149 - 140\cos\theta.$$

Since $NP \perp AN$, from the Pythagorean Theorem on triangle ANP, we have

$$AP^2 = AN^2 + NP^2 = AN^2 + MC^2 = 298.$$

So, P lies on the circle centered at A with radius $\sqrt{298}$, which bounds a region with area $298\pi \implies \boxed{298}$.

TR 12: For a positive integer n, let S_n denote the sum of all positive integers m such that $\frac{n}{m-3}$ is a positive integer. Suppose k is a positive integer such that $S_{2k} = 1650$, $S_{4k} = 3492$, and $S_{8k} = 7158$. Find S_k.

Answer: $\boxed{720}$

Solution: Note that $\frac{n}{m-3}$ is a positive integer if and only if $m - 3$ is a positive divisor of n. So, $S_n = \sigma(n) + 3\tau(n)$, where $\sigma(n)$ is the sum of the divisors of n and $\tau(n)$ is the number of divisors of n. Let $k = 2^e q$, where q is odd, and let $A = \sigma(q)$ and $T = \tau(q)$. Then,

$$S_{2^j k} = \left(2^{e+j+1} - 1\right) A + 3(e + j + 1)T.$$

Subtracting consecutive equations, we have

$$S_{4k} - S_{2k} = 2^{e+2}A + 3T = 1842, \quad \text{and}$$
$$S_{8k} - S_{4k} = 2^{e+3}A + 3T = 3666.$$

Subtracting these, we have $2^{e+2}A = 1824$, so $3T = 18 \implies T = 6$. Substituting into $S_{2k} = 1650$, we have

$$\left(2^{e+2} - 1\right) A + 18(e + 2) = 1650 \implies 1824 - A + 18e + 36 = 1650 \implies A = 210 + 18e.$$

Since $2^{e+2}A = 1824 = 2^5 \cdot 57$, checking $e = 0, 1, 2, 3$ gives $e = 1$ and $A = 228$. (For example, $q = 147$ works.) So,

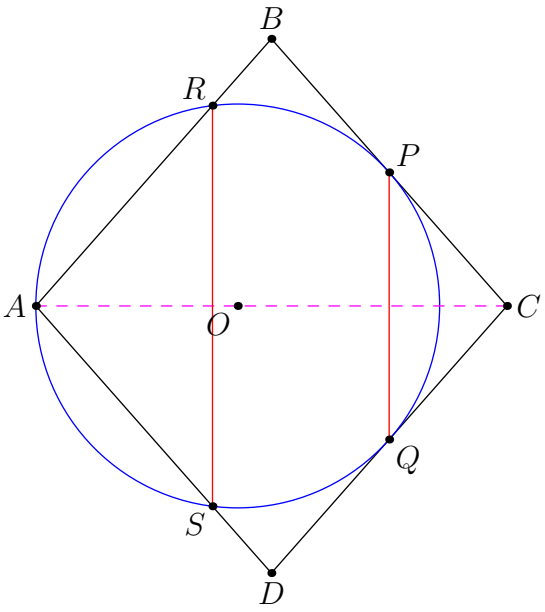
$$S_k = (2^{e+1} - 1)A + 3(e + 1)T = 3 \cdot 228 + 3 \cdot 2 \cdot 6 = \boxed{720}.$$

TR 13: Let $ABCD$ be a rhombus. Circle ω passes through A and is tangent to sides BC and CD at points P and Q, respectively. Points R and S distinct from A are the intersections of ω with sides AB and DA, respectively. Given that $PQ = 2$ and $RS = 3$, the square of the area of $ABCD$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive

integers. Find $m+n$.

Answer: $\boxed{457}$

Solution:



Let O and r be the center and radius of ω, and let $\alpha = \angle BAC = \angle BCA$. Since ω is tangent to BC and CD, O lies on the angle bisector of $\angle BCD$, which is AC. Since $OA = r$, we have $OC = AC - r$, and since the distance from O to BC is r,

$$(AC - r)\sin\alpha = r.$$

Since AC contains a diameter of ω and $\angle RAC = \alpha$, we have $AR = AS = 2r\cos\alpha$, so $RS = 2AR\sin\alpha = 4r\sin\alpha\cos\alpha$. In addition, $CP = CQ = OC\cos\alpha$, so $PQ = 2CP\sin\alpha = 2(AC - r)\sin\alpha\cos\alpha$. Dividing, we have

$$\frac{RS}{PQ} = \frac{2r}{AC - r} = \frac{3}{2} \implies AC = \frac{7r}{3}.$$

Substituting into $(AC - r)\sin\alpha = r$, we have $\frac{4r}{3}\sin\alpha = r$, so $\sin\alpha = \frac{3}{4}$ and $\cos\alpha = \frac{\sqrt{7}}{4}$. So,

$$PQ = 2 \cdot \frac{4r}{3} \cdot \frac{3}{4} \cdot \frac{\sqrt{7}}{4} = \frac{r\sqrt{7}}{2} = 2 \implies r = \frac{4}{\sqrt{7}},$$

meaning $AC = \frac{4\sqrt{7}}{3}$ and $BD = AC\tan\alpha = \frac{4\sqrt{7}}{3} \cdot \frac{3}{\sqrt{7}} = 4$. Thus, the area of $ABCD$ is $\frac{AC \cdot BD}{2} = \frac{8\sqrt{7}}{3}$, and its square is

$$\frac{448}{9} \implies 448 + 9 = \boxed{457}.$$

TR 14: Determine the number of ways we can place a number in each cell of a 32×32 grid such that the following three conditions are met:

- Every positive integer less than or equal to 32^2 appears in the grid;

- For any 32 cells of the grid such that no two lie in the same row or column, the sum of the numbers in those cells is always the same;
- The numbers in each row increase from left to right, and the numbers in each column increase from bottom to top.

Answer: $\boxed{252}$

Solution: Let $a_{i,j}$ be the number in the ith row from the bottom and the jth column from the left. Swapping two cells between rows i, k and columns j, l in a set of 32 cells, we have $a_{i,j} + a_{k,l} = a_{i,l} + a_{k,j}$ for all i, j, k, l. Since $a_{1,1} = 1$, this means

$$a_{i,j} - 1 = (a_{i,1} - 1) + (a_{1,j} - 1) = r_i + c_j,$$

where $r_i = a_{i,1} - 1$ and $c_j = a_{1,j} - 1$. Let $R = \{r_i\}$ and $C = \{c_j\}$, which both contain 0 and have 32 elements. Then, every integer from 0 to 1023 can be written uniquely as $r + c$ with $r \in R$ and $c \in C$. Conversely, each such pair (R, C) gives exactly one grid, since the conditions force the rows and columns to be in increasing order. Now, it is well known (and can be shown by induction) that if every integer from 0 to $N - 1$ can be written uniquely as $x + y$ with $x \in X$ and $y \in Y$, where $1 \in X$ and m is the smallest positive element of Y, then

$$X = \{0, 1, \ldots, m - 1\} + mX' \quad \text{and} \quad Y = mY',$$

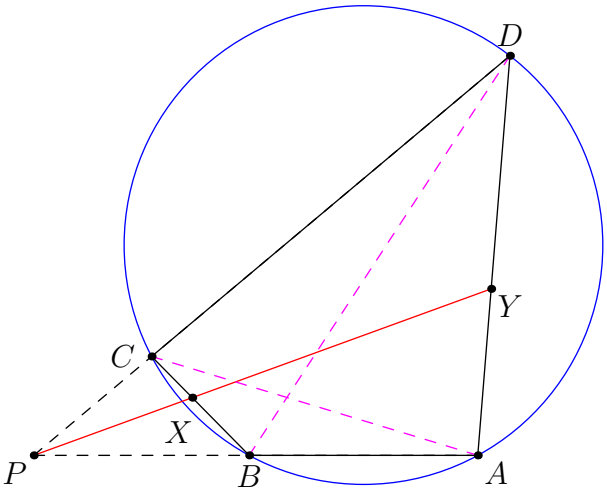
where every integer from 0 to $\frac{N}{m} - 1$ can be written uniquely as $x' + y'$ with $x' \in X'$ and $y' \in Y'$. Repeating this, every pair (R, C) corresponds to writing $1024 = 2^{10}$ as a product of powers of 2 greater than 1, where the factors are alternately assigned to R and C, and the factors assigned to each set multiply to $32 = 2^5$. If there are j factors assigned to each set, then there are $\binom{4}{j-1}^2$ ways to choose the exponents and 2 ways to choose which set gets the first factor. If one set has one more factor than the other, then that set gets the first and last factors, and there are $2\sum_j \binom{4}{j}\binom{4}{j-1}$ ways. So, the answer is

$$2\sum_{j=1}^{5}\binom{4}{j-1}^2 + 2\sum_{j=1}^{4}\binom{4}{j}\binom{4}{j-1} = 2\binom{8}{4} + 2\binom{8}{5} = 140 + 112 = \boxed{252}.$$

TR 15: Let $ABCD$ be a cyclic quadrilateral. Lines AB and CD intersect at point P. The angle bisector of $\angle BPC$ intersects sides BC and AD at points X and Y, respectively. Given that $AC = 5$, $BD = 7$, and $XY = \frac{14}{3}$, the area of $ABCD$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{89}$

Solution:



Since $ABCD$ is cyclic, from power of a point, $PA \cdot PB = PC \cdot PD$, so triangles PAC and PDB are similar with ratio $\lambda = \frac{AC}{BD} = \frac{5}{7}$. Let $PB = b$ and $PD = d$, so $PA = \lambda d$ and $PC = \lambda b$. Let P be the origin, and let u and v be the unit vectors in the directions of $\overrightarrow{PB}$ and $\overrightarrow{PC}$, respectively. From the angle bisector theorem on triangles PBC and PAD, we have

$$X = \frac{PC \cdot B + PB \cdot C}{PB + PC} = \frac{\lambda b}{1+\lambda}(u+v) \quad \text{and} \quad Y = \frac{PD \cdot A + PA \cdot D}{PA + PD} = \frac{\lambda d}{1+\lambda}(u+v),$$

so $XY = \frac{\lambda}{1+\lambda}|d-b||u+v|$. Now, since $u+v \perp v-u$, we have

$$\overrightarrow{BD} = dv - bu = \frac{d-b}{2}(u+v) + \frac{d+b}{2}(v-u),$$

so the projection of BD onto line XY has length $\frac{|d-b||u+v|}{2}$. In the same manner, $\overrightarrow{AC} = \lambda(bv - du)$ has a projection of length $\frac{\lambda|d-b||u+v|}{2}$. Since the projections of AC and BD are proportional to their lengths, line XY makes the same angle ψ with both diagonals, so the diagonals meet at an angle of 2ψ. This gives

$$XY = \frac{2\lambda}{1+\lambda} \cdot BD \cos\psi = \frac{2 \cdot AC \cdot BD}{AC + BD} \cos\psi \implies \frac{14}{3} = \frac{70}{12}\cos\psi \implies \cos\psi = \frac{4}{5}.$$

So, $\sin(2\psi) = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$, and the area of $ABCD$ is

$$\frac{1}{2} \cdot AC \cdot BD \cdot \sin(2\psi) = \frac{1}{2} \cdot 5 \cdot 7 \cdot \frac{24}{25} = \frac{84}{5} \implies 84 + 5 = \boxed{89}.$$